Input: design d, attacker profile a, legitimate-traffic specification ℓ, seed z, attack minutes T, warm-up minutes W
Output: the run record R

$(\mathrm{rng}_\ell, \mathrm{rng}_w, \mathrm{rng}_p) \leftarrow$ STREAMS(z) ▷ users, attacker, challenge outcomes; the same z offers the same users to every scenario
$P \leftarrow$ BUILDPIPELINE(d) on the in-memory store with vendor fakes; SEEDRETURNINGUSERS$(P, 0.2)$
$E \leftarrow \emptyset$; leaked, offered $\leftarrow$ empty arrays ▷ E: timed events (receipts, code entries, resends)
for $m \leftarrow -W$ **to** $T - 1$ **do**
  $n_\ell \sim \mathrm{Poisson}(\ell.\mathrm{rate})$ from rng_ℓ; $n_a \sim \mathrm{Poisson}(a.\mathrm{rate})$ from rng_w if $m \geq 0$, else 0
  $A \leftarrow$ the $n_\ell + n_a$ arrivals with times drawn inside minute m, sorted by time
  for all $(t, \mathrm{kind}) \in A$ **do**
    APPLYEVENTS(E, t); clock $\leftarrow t$
    $q \leftarrow$ DRAWREQUEST(kind) ▷ identity, address, number, CAPTCHA scores, delays and whether the person would verify, all drawn in advance
    $r \leftarrow P$.PROCESSREQUEST(q) ▷ Algorithm 2
    if r = challenge **and** q solves it **then** $r \leftarrow P$.PROCESSREQUEST$(q \,\|\, \mathrm{solution})$
    if r sent an SMS **then**
      push onto E a receipt at $t +$ queue $+$ delivery and, if q verifies, a code entry after its delay
  APPLYEVENTS$(E, \mathrm{end}(m))$; P.RESOLVETIMEOUTS(); P.ADAPTIVEJOB() at its cadence
  if $m \geq 0$ **then** leaked$[m] \leftarrow$ SMS sent to a in minute m; offered$[m] \leftarrow n_a$
APPLYEVENTS$(E, \mathrm{horizon})$ ▷ drain: receipt grace, resolution timeout, the slowest entry and a resend
return $R \leftarrow$ (leaked, offered, containment; each user's funnel by cohort; verdict events; the attacker's verifications, challenges solved and bill; d, a, ℓ, z, the workload digest and the code hash)